

MYEONGSEOK RYU

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Myeongseok Ryu is under Ph.D. program. (compiled on September 28, 2026)

RESEARCH INTERESTS

Control Theory

Adaptive Control, Optimal Control

Neural Network-based Control

Neuro-Adaptive Control, Reinforcement Learning

Contraction Theory

Vehicle Dynamics and Control

Active Rear Steering (ARS) Control

PROFILE & LOGOS



EDUCATION

Korea Advanced Institute of Science and Technology (KAIST), Korea

CCS Graduate School of Mobility

Ph.D. of Science in Mobility Engineering

September 2025 – Present

- Supervisor: Prof. Kyunghwan Choi, KAIST

Gwangju Institute of Science and Technology (GIST), Korea, (Withdrew for further studies)

School of Mechanical Engineering

Ph.D. of Science in Mechanical Engineering

March 2025 – May 2025

Gwangju Institute of Science and Technology (GIST), Korea

School of Mechanical Engineering

Master of Science in Mechanical Engineering

March 2023 – February 2025

- Thesis: Constrained Optimization-Based Neuro-Adaptive Control (CoNAC) for Euler-Lagrange Systems
- Supervisor: Prof. Kyunghwan Choi, GIST

Incheon National University (INU), Korea

Department of Mechanical Engineering

Bachelor of Engineering

March 2017 – February 2023

PROFESSIONAL EXPERIENCE

Korea Advanced Institute of Science and Technology (KAIST), Korea

Researcher, MIC Lab, CCS Graduate School of Mobility

– Research on Neural Network-based Control for Mobility Systems

May 2025 – August 2025

RESEARCH PROJECTS

Development of a Rear-Wheel Steering (RWS) Control Algorithm for Improved Vehicle Handling Performance and Tunability (Phase II)

September 2026 – August 2027

Sponsor: Hyundai Motor Company (HMC)

Role: Lead Researcher

– Research on Active Rear Steering (ARS) Control for Vehicle Lateral Dynamics

Development of a Rear-Wheel Steering (RWS) Control Algorithm for Improved Vehicle Handling Performance and Tunability (Phase I)

August 2025 – July 2026

Sponsor: Hyundai Motor Company (HMC)

Role: Lead Researcher

– Research on Active Rear Steering (ARS) Control for Vehicle Lateral Dynamics

SKILLS

Languages: Korean, English
Programming: Matlab/Simulink, Python, C/C++
Implementation: **Simulation** CarMaker, ROS
Others Git, LaTeX, Jekyll

PUBLICATIONS

Papers Under Review

3. Constrained Optimization-Based Neuro-Adaptive Control (CONAC) for Unknown Systems Under Multiple Convex Input Constraints
Myeongseok Ryu, Donghwa Hong, Kyunghwan Choi*
Withheld during double-blind review, 2026
2. Model Predictive Control for Active Rear Steering Without Sideslip-Angle Feedback
Myeongseok Ryu, Soobin Hwang, Kyunghwan Choi*
American Control Conference (ACC), 2027
1. Physics-Constrained Kalman Filtering for Wheel Normal Load Estimation Using Suspension and IMU Measurements
Angus Siegloff, Myeongseok Ryu, Kyunghwan Choi*
IEEE Transactions on Vehicle Technology, 2026

International Conference Papers

7. Constrained Optimization-Based Yaw-Rate Reference Map Generation for Active Rear Steering Control of Four-Wheel-Steering Vehicles
Myeongseok Ryu, Donghyun Hwang, Youngsik Yoon, Kyunghwan Choi*
International Conference on Control, Automation and Systems (ICCAS), (accepted, in press), 2026
6. Vector-Space Optimization Framework Based on Projected Contraction Condition for Control Design with Input Saturation
Myeongseok Ryu, Kyunghwan Choi*, Sesun You
International Federation of Automatic Control (IFAC) World Congress, (accepted, in press), 2026
5. All-Wheel Steering Vehicle Control Based on Contraction Theory with Neural Network
Myeongseok Ryu, Kyunghwan Choi*
IEEE International Conference on Advanced Motion Control (AMC), pp. 1-6, 2026
4. Physics-Informed Online Learning of Flux Linkage Model for Synchronous Machine
Seunghun Jang, Myeongseok Ryu, Kyunghwan Choi*
Annual Conference of the IEEE Industrial Electronics Society (IECON), pp. 1-7, 2025
3. Constrained Optimization-Based Neuro-Adaptive Control (CONAC) for Synchronous Machine Drives Under Voltage Constraints
Myeongseok Ryu, Niklas Monzen, Pascal Seitter, Kyunghwan Choi, Christoph M. Hackl*
Annual Conference of the IEEE Industrial Electronics Society (IECON), pp. 1-7, 2025
2. Imposing a Weight Norm Constraint for Neuro-Adaptive Control
Myeongseok Ryu, Jiyun Kim, Kyunghwan Choi*
European Control Conference (ECC), pp. 380-385, 2025
1. A Comparative Study of Reinforcement Learning and Analytical Methods for Optimal Control
Myeongseok Ryu, Junseo Ha, Minji Kim, Kyunghwan Choi*
International Workshop on Intelligent Systems (IWIS), pp. 1-5, 2023

Domestic Conference Papers

4. Constrained Optimization-Based Performance Improvement Analysis of Vehicle Lateral Control Using All-Wheel Steering
Myeongseok Ryu, Kyunghwan Choi*
제어로봇시스템학회 (ICROS), (accepted, in press), 2026
3. Approximation-based Steering Controller with Deep Neural Network
Myeongseok Ryu, Kyunghwan Choi*
제어로봇시스템학회 (ICROS), pp. 884-885, 2024
2. Integrated Motion Control of Four in-Wheel Motor Actuated Vehicles Considering Path Tracking, Ride Comfort, and Energy Efficiency
Myeongseok Ryu, Kyunghwan Choi*
한국자동차공학회 추계학술대회 (KSAE), pp. 490, 2023
1. Data-driven Modeling of Model Residuals for Linear Model Predictive Control of Nonlinear Systems
Myeongseok Ryu, Kyunghwan Choi*
제어로봇시스템학회 (ICROS), pp. 837-838, 2023

Preprint Papers

1. CNN-based End-to-End Adaptive Controller with Stability Guarantees
Myeongseok Ryu, Kyunghwan Choi*
Arxiv, 2024

GRANTS AND AWARDS

Best Presentation Paper Award <i>IEEE International Workshop on Intelligent Systems (IWIS)</i>	<i>July 2025</i>
Student Support <i>European Control Association (EUCA)</i>	<i>June 2025</i> 400 EUR
Research Support <i>Graduate International Research Experience Fellowship (GIST-IREF)</i>	<i>October 2024</i> 16 million KRW (approx. 12,000 USD)
Best Paper Award <i>Institute of Control, Robotics and Systems (ICROS)</i>	<i>June 2023</i>
INU MATLAB Cody Challenge <i>Top Prize</i>	<i>June 2021</i>